ΕΡΓΑΣΙΑ ΘΗΝΟΥ

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exercise set 1.3

(33)

$$\begin{pmatrix} 4 & 3 & 3 \\ 2 & 1 & 0 \\ 4 & 4 & 2 \end{pmatrix}$$

a) $y = \begin{pmatrix} 4 & 3 & 3 \\ 2 & 1 & 0 \\ 4 & 4 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \\ 1.50 \end{pmatrix}$

b) $y = \begin{pmatrix} 8+3+4.5 \\ 4+1 \\ 8+4+3 \end{pmatrix} = \begin{pmatrix} 15.5 \\ 5 \\ 15 \end{pmatrix}$

c) $\begin{pmatrix} 4 & 3 & 3 \\ 2 & 2 & 2 \\ 4 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 1.50 \end{pmatrix} = \begin{pmatrix} 15.5 \\ 9 \\ 15 \end{pmatrix}$

(24)

b) $Ax = b$

$LUx = b$ (1)

$$L = \begin{pmatrix} 2 & 0 & 0 \\ 4 & 1 & 0 \\ -3 & -2 & 3 \end{pmatrix}, \quad U = \begin{pmatrix} 3 & -5 & 2 \\ 0 & 4 & 1 \\ 0 & 0 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix}$$

$Ly = b$

$$\begin{bmatrix} 2 & 0 & 0 \\ 4 & 1 & 0 \\ -3 & -2 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 4 \\ -5 \\ 2 \end{bmatrix} \Rightarrow 2y_1 = 4 \Rightarrow y_1 = 2$$

$4y_1 + y_2 = -5 \Rightarrow y_2 = -13$

$-3y_1 - 2y_2 + 3y_3 = 2 \Rightarrow y_3 = -6$

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$$y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 2 \\ -13 \\ -6 \end{pmatrix}$$

$Ux = y$

$$\begin{pmatrix} 3 & -5 & 2 \\ 0 & 4 & 1 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ -13 \\ -6 \end{pmatrix} \Rightarrow \begin{matrix} 3x_1 - 5x_2 + 2x_3 = 2 \Rightarrow x_1 = -\frac{3}{2} \\ 4x_2 + x_3 = -13 \Rightarrow x_2 = -\frac{5}{2} \\ 2x_3 = -6 \Rightarrow x_3 = -3 \end{matrix}$$

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -3/2 \\ -5/2 \\ -3 \end{pmatrix}$$

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exercise set 2.1

(29)

$$\begin{vmatrix} + & - & + \\ x & y & 1 \\ a_1 & b_1 & 1 \\ a_2 & b_2 & 1 \end{vmatrix} = x \begin{vmatrix} b_1 & 1 \\ b_2 & 1 \end{vmatrix} - y \begin{vmatrix} a_1 & 1 \\ a_2 & 1 \end{vmatrix} + 1 \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

$$= x(b_1 - b_2) - y(a_1 - a_2) + b_2 a_1 - a_2 b_1$$

$$\begin{vmatrix} x & y & 1 \\ a_1 & b_1 & 1 \\ a_2 & b_2 & 1 \end{vmatrix} = x(b_1 - b_2) - y(a_1 - a_2) + b_2 a_1 - a_2 b_1 \quad (1)$$

$$\frac{y - b_2}{b_2 - b_1} = \frac{x - a_2}{a_2 - a_1} \Rightarrow x(b_1 - b_2) - y(a_1 - a_2) + b_2 a_1 - a_2 b_1 = 0$$

$$\Rightarrow \begin{vmatrix} x & y & 1 \\ a_1 & b_1 & 1 \\ a_2 & b_2 & 1 \end{vmatrix} = 0$$

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(31)

a) $A = \begin{pmatrix} A11 & A12 \\ 0 & A22 \end{pmatrix}$ $A11 = \begin{pmatrix} 2 & -1 \\ 4 & 3 \end{pmatrix}$, $A12 = \begin{pmatrix} 2 & 5 & 6 \\ -1 & 3 & 4 \end{pmatrix}$

$detA = det(A11)\, det(A22)$ $A22 = \begin{pmatrix} 1 & 3 & 5 \\ -2 & 6 & 2 \\ 3 & 5 & 2 \end{pmatrix}$

$det(A11) = 2 \cdot 3 - [4(-1)] = 10$

$det(A22) = -1\begin{pmatrix} 6 & 2 \\ 5 & 2 \end{pmatrix} - 3\begin{pmatrix} -2 & 2 \\ 3 & 2 \end{pmatrix} + 5\begin{pmatrix} -2 & 6 \\ 3 & 5 \end{pmatrix} = -108$

b) $det|A| = 2\begin{vmatrix} 3 & -1 & 3 & 4 \\ 0 & 1 & 3 & 5 \\ 0 & -2 & 6 & 2 \\ 0 & 3 & 5 & 2 \end{vmatrix} \begin{vmatrix} -4 \end{vmatrix} \begin{vmatrix} -1 & 2 & 5 & 6 \\ 0 & 1 & 3 & 5 \\ 0 & -2 & 6 & 2 \\ 0 & 3 & 5 & 2 \end{vmatrix} = -1080$

exercise set 1.4

(23)

$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$

$Ax = I$

$detA = 1|1-0| - 0|1-0| + 1|1-0| = 2 \neq 0$

$x = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

ΠΡΩΤΟΠΟΡΙΑ ΠΑΝΕΠΙΣΤΗΜΙΟ ΠΑΤΡΩΝ ΟΥΚ ΕΛΑΤΤΩ ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ

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$a+g=1 \quad b+h=0 \quad c+i=0 \Rightarrow a=1/2,\ b=1/2,\ c=-1/2$

$a+d=0 \quad b+e=1 \quad c+f=0 \quad d=-1/2,\ e=1/2,\ f=1/2$

$d+g=0 \quad e+h=0 \quad f+i=1 \quad g=1/2,\ h=-1/2,\ i=1/2$

$$x=\begin{pmatrix} 1/2 & 1/2 & -1/2 \\ -1/2 & 1/2 & 1/2 \\ 1/2 & -1/2 & 1/2 \end{pmatrix}$$

exercise set 1.2

(21)

$x=\sin\alpha,\ y=\cos\beta,\ z=\tan\gamma$

$$\begin{cases} x+2y+3z=0 \\ 2x+5y+3z=0 \\ -x-5y+5z=0 \end{cases}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 5 & 3 & 0 \\ -1 & -5 & 5 & 0 \end{array}\right) \xrightarrow{R2:R1-2R} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & -3 & 8 & 0 \end{array}\right)$$

$$\xrightarrow{R3:R3+3R2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & -1 & -3 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right) \xrightarrow{R3:-R3} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$$\xrightarrow{R1:R1-3R3} \left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) \xrightarrow{R1:R1-2R2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$x=0,\ y=0,\ z=0$

$\Rightarrow \sin\alpha=0,\ \cos\beta=0,\ \tan\gamma=0$

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(22)

$\lambda \neq 3$:

$y = -(\lambda - 5)x$

$y = \frac{-x}{(\lambda - 3)}$

$(\lambda - 3) = \frac{1}{(\lambda - 3)} \Rightarrow (\lambda - 3)^2 = 1 \Rightarrow (\lambda - 3 + 1)(\lambda - 3 - 1) = 0 \Rightarrow (\lambda - 2)(\lambda - 4) = 0$

$\lambda = 2, \lambda = 4$

exercise set 1.5

(7)

a) $[A|I_3] = \left[\begin{array}{ccc|ccc} 3 & 4 & -1 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 2 & 5 & 4 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R1 <-> R2} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 3 & 4 & -1 & 1 & 0 & 0 \\ 2 & 5 & 4 & 0 & 0 & 1 \end{array}\right]$

$\xrightarrow[-2R1]{-3R1} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 4 & -10 & 1 & -3 & 0 \\ 0 & 5 & -10 & 0 & -2 & 1 \end{array}\right] \xrightarrow{-R3} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 & -1 & -1 \\ 0 & 5 & -10 & 0 & -2 & 1 \end{array}\right]$

$\xrightarrow[+5R2]{-R2} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & -1 & 0 & -1 & 1 & 1 \\ 0 & 0 & -10 & 0 & -2 & 1 \end{array}\right] \xrightarrow{\cdot \frac{-R3}{10}} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 1 & 1 \\ 0 & 0 & 1 & -0,5 & 0,7 & 0,4 \end{array}\right]$

$\xrightarrow{-3R3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1,5 & -1,1 & -1,2 \\ 0 & 1 & 0 & -1 & 1 & 1 \\ 0 & 0 & 1 & -0,5 & 0,7 & 0,4 \end{array}\right]$

$A^{-1} = \begin{bmatrix} 1,5 & -1,1 & -1,2 \\ -1 & 1 & 1 \\ -0,5 & 0,7 & 0,4 \end{bmatrix}$

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exercise set 3.1

(21)

a) Λάθος
b) Λάθος
c) Λάθος
d) Σωστό
e) Σωστό
f) Λάθος

exercise set 2.1

(16)

$7x_1 - 2x_2 = 3$

$3x_1 + x_2 = 5$

$A = \begin{pmatrix} 7 & -2 \\ 3 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$

$A1 = \begin{pmatrix} 3 & -2 \\ 5 & 1 \end{pmatrix}$, $A2 \begin{pmatrix} 7 & 3 \\ 3 & 5 \end{pmatrix}$

$\det A = \begin{vmatrix} 7 & -2 \\ 3 & 1 \end{vmatrix} = 13$

$\det A1 = \begin{vmatrix} 3 & -2 \\ 5 & 1 \end{vmatrix} = 13$

$\det A2 = \begin{vmatrix} 7 & 3 \\ 3 & 5 \end{vmatrix} = 26$

$x_1 = \frac{\det A1}{\det A} = \frac{13}{13} = 1$

$x_2 = \frac{\det A2}{\det A} = \frac{26}{13} = 2$

$\Downarrow$

$(x_1, x_2) = (1, 2)$

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exercise set 3.3

(21)

$$\cos\alpha = \frac{a}{\|v\|}$$

$$\cos\alpha = \frac{v\cdot i}{\|v\|} \Rightarrow \cos\alpha = \frac{a}{\|v\|}$$

$$\cos\beta = \frac{v\cdot j}{\|v\|} = \frac{b}{\|v\|}$$

$$\cos\gamma = \frac{v\cdot k}{\|v\|} = \frac{c}{\|v\|}$$

$$\frac{v}{\|v\|} = \left(\frac{a}{\|v\|}, \frac{b}{\|v\|}, \frac{c}{\|v\|}\right) = (\cos\alpha, \cos\beta, \cos\gamma)$$

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

$$\|v\| = \sqrt{\cos^2\alpha + \cos^2\beta + \cos^2\gamma} = 1$$

(25)

u, v ορθογώνια αν $u\cdot v = 0$

Έστω v ορθογώνιο διανύσματα στο w_1, w_2

$\Rightarrow v\cdot w_1 = 0$, $v\cdot w_2 = 0$

$$v(k_1w_1 + u_2w_2) = v(k_1w_1) + v(k_2w_2)$$
$$= k_1(vw_1) + k_2(vw_2)$$
$$= k_1(0) + k_2(0)$$
$$= 0$$

Άρα το v είναι ορθογώνιο στο $u_1w_1 + u_2w_2$ για όλες τις γραμμικές u_1, u_2.

exercise set 5.1

(18)

$$A_i = \begin{pmatrix} a_i & 1 \\ 1 & b_i \end{pmatrix} \quad i = 1, 2$$

$$A_1 + A_2 = \begin{pmatrix} a_1+a_2 & 1 \\ 1 & b_1+b_2 \end{pmatrix}$$

$$A_1 + (A_2 + A_3) = \begin{pmatrix} a_1 & 1 \\ 1 & b_1 \end{pmatrix} + \left[\begin{pmatrix} a_2 & 1 \\ 1 & b_2 \end{pmatrix} + \begin{pmatrix} a_3 & 1 \\ 1 & b_3 \end{pmatrix}\right]$$

$$= \begin{pmatrix} a_1 + (a_2+a_3) & 1 \\ 1 & b_1 + (b_2+b_3) \end{pmatrix} = \begin{pmatrix} (a_1+a_2)+a_3 & 1 \\ 1 & (b_1+b_2)+b_3 \end{pmatrix} *$$

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$$* \quad = \left[\begin{pmatrix} a_1 & 1 \\ 1 & b_1 \end{pmatrix} + \begin{pmatrix} a_2 & 1 \\ 1 & b_2 \end{pmatrix} \right] + \begin{pmatrix} a_3 & 1 \\ 1 & b_3 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in V \;, \quad \begin{pmatrix} a & 1 \\ 1 & b \end{pmatrix} \in V$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} a & 1 \\ 1 & b \end{pmatrix} = \begin{pmatrix} a & 1 \\ 1 & b \end{pmatrix}$$

$$A = \begin{pmatrix} a & 1 \\ 1 & b \end{pmatrix} \;, \quad A' = \begin{pmatrix} -a & 1 \\ 1 & -b \end{pmatrix} \Bigg\} \begin{pmatrix} a & 1 \\ 1 & b \end{pmatrix} + \begin{pmatrix} -a & 1 \\ 1 & -b \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$A = \begin{pmatrix} a & 1 \\ 1 & b \end{pmatrix} \;, \quad kA \begin{pmatrix} ka & 1 \\ 1 & kb \end{pmatrix}$$

$$k(A_1 + A_2) = k \begin{pmatrix} a_1 + a_2 & 1 \\ 1 & b_1 + b_2 \end{pmatrix} = \begin{pmatrix} ka_1 & 1 \\ 1 & ka_2 \end{pmatrix} \begin{pmatrix} ka_2 & 1 \\ 1 & kb_2 \end{pmatrix} = kA_1 + kA_2$$

$$(k+m)A = \begin{pmatrix} (k+m)a & 1 \\ 1 & (k+m)b \end{pmatrix} = kA + mA$$

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exercise set 5.5

(14)

$$A=\begin{pmatrix}0&1&0\\1&0&0\\0&0&0\end{pmatrix}$$

$$Ax=0_m \Rightarrow \begin{pmatrix}0&1&0\\1&0&0\\0&0&0\end{pmatrix}\begin{pmatrix}x1\\x2\\x3\end{pmatrix}=\begin{pmatrix}x2\\x1\\0\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$$

$x1=0$, $x2=0$

$$\text{Μη Σειραχώρος}=\{(x1,x2,x3)=(0,0,x3)\ x\in\mathbb{R}\}$$

$$Ax=\begin{pmatrix}0&1&0\\1&0&0\\0&0&0\end{pmatrix}\begin{pmatrix}x1\\x2\\x3\end{pmatrix}=c_1x_1+c_2x_2+c_3x_3$$

$$=\begin{pmatrix}0\\1\\0\end{pmatrix}x1+\begin{pmatrix}1\\0\\0\end{pmatrix}x2+\begin{pmatrix}0\\0\\0\end{pmatrix}x3=\begin{pmatrix}x2\\x1\\0\end{pmatrix}$$

$$\dim(LS)=\left\{x=\begin{pmatrix}x2\\x1\\0\end{pmatrix}\right\}_{x\in\mathbb{R}^2}$$

(17)

a) $A=\begin{pmatrix}\alpha&\beta\\\gamma&\delta\end{pmatrix}$ $\quad \text{Μη Σειραχώρος}=\{Ax=0\}$

$3x=5y \Rightarrow x=\frac{5y}{3}$

$$Ax=0$$

$$\Rightarrow\begin{pmatrix}\alpha&\beta\\\gamma&\delta\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}\Rightarrow\begin{pmatrix}\alpha x&\beta y\\\gamma x&\delta y\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}$$

$\alpha x+\beta y=0 \Rightarrow \alpha=3$, $\beta=-5$

$\gamma x+\delta y=0 \Rightarrow \gamma=3$, $\delta=-5$

$\text{Μη Σειραχώρος}=\{(x,y)=(0,0)\}$

$B\begin{pmatrix}1&0\\0&5\end{pmatrix}$, $Bx=0$

$$\begin{pmatrix}1&0\\0&5\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}\Rightarrow\begin{pmatrix}x\\5y\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}$$

$x=0$, $y=0$ $\quad \text{Μη Σειραχώρος}=\{(x,y)=(0,0)\}$

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$\Gamma = \begin{pmatrix} 6 & 2 \\ 3 & 1 \end{pmatrix}$, $\Gamma x = 0$

$\begin{pmatrix} 6 & 2 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 6x+2y \\ 3x+y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\begin{matrix} 6x+2y=0 \\ 3x+y=0 \end{matrix}\Big\} y=-3x$

$3x-5y=0 \Rightarrow A = \begin{bmatrix} 3 & -5 \\ 3 & -5 \end{bmatrix}$ $\begin{matrix} \alpha=\gamma=3 \\ \beta=\delta=-5 \end{matrix}$

β) $A = \begin{bmatrix} 1 & 4 \\ 0 & 5 \end{bmatrix}$, $Ax=0$

$\begin{bmatrix} 1 & 4 \\ 0 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x+4y \\ 5y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\begin{matrix} x=0 \\ y=0 \end{matrix}$

$D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \emptyset$

$Dx=0 \Rightarrow \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow$ $\begin{matrix} 0x=0 \\ 0y=0 \end{matrix}\Big\} 0=0 \Rightarrow (x,y) \in \mathbb{R}^2$

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Π. Χριστοδουλίδης 6.4

(3) a) $\{u_1=(1,-1,1), u_2=(2,0,1), u_3=(0,1,1)\}$

$v_1=\frac{1}{\|w\|}$ $w=\frac{1}{\sqrt{1^2+(-1)^2+1^2}}=\frac{1}{\sqrt{3}} \Rightarrow w=\frac{1}{\sqrt{3}}(1,-1,1)$

$w_2=u_2-(u_2\cdot v_1)v_1=(2,0,1)-\left[(2,0,1)\frac{1}{\sqrt{3}}(1,-1,1)\right]\cdot\frac{1}{\sqrt{3}}(1,-1,1)$

$=(2,0,1)-\left(\frac{3}{\sqrt{3}}\cdot\frac{1}{\sqrt{3}}(1,-1,1)\right)=(2,0,1)-(1,-1,1)=(1,1,0)$

$v_2=\frac{1}{\|w_2\|}$ $w_2=\frac{1}{\sqrt{1^2+1^2+0^2}} \Rightarrow w_2=\frac{1}{\sqrt{2}}(1,1,0)$

$w_3=u_3-(u_3\cdot v_1)\cdot v_1-(u_3\cdot v_2)v_2$

$u_3\cdot v_1=(0,1,1)\frac{1}{\sqrt{3}}(1,-1,1)=0$

$u_3\cdot v_2=(0,1,1)\frac{1}{\sqrt{2}}(1,1,0)=\frac{1}{\sqrt{2}}$

$w_3=(0,1,1)-(0\cdot v_1)-\frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}(1,1,0)=\left(-\frac{1}{2},\frac{1}{2},1\right)$

$v_3=\frac{1}{\|w_3\|}$ $w_3=\frac{1}{\sqrt{\left(-\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^2+1^2}}\cdot\left(-\frac{1}{2},\frac{1}{2},1\right)=\left(\frac{-1}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}}\right)$

Ορθοκανονική βάση $\Rightarrow v_1=\frac{1}{\sqrt{3}}(1,-1,1)$

$v_2=\frac{1}{\sqrt{2}}(1,1,0)$

$v_3=\frac{1}{\sqrt{6}}(-1,1,2)$

ΠΛΗΡΟΦΟΡΙΑ ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ

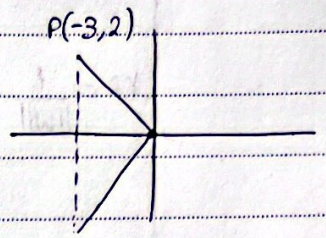
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Κεφάλαιο 7

(9) a) P(-3,2)

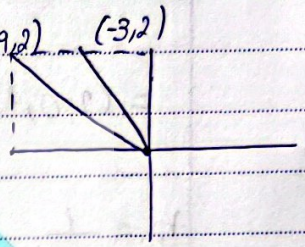
T1 = Reflection in the x-axis

$$M = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} -3 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ -2 \end{pmatrix}$$



T2 = Stretch by a factor of 3 to Ox

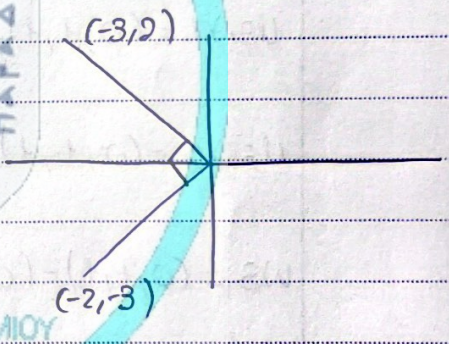
$$M = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -3 \\ 2 \end{pmatrix} = \begin{pmatrix} -9 \\ 2 \end{pmatrix}$$



T3 = Rotation by 90° about the origin

$$M = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} -3 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}$$



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Κεφάλαιο 8

Ασκήσεις 8.2

(1) α) $A = \begin{pmatrix} 1 & -2 \\ 4 & 3 \end{pmatrix}$ $e(\lambda) = \det(A - \lambda I)$

$$A - \lambda I = \begin{pmatrix} 1 & -2 \\ 4 & 3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 1-\lambda & -2 \\ 4 & 3-\lambda \end{pmatrix}$$

$$e(\lambda) = \det(A - \lambda I) = (1-\lambda)(3-\lambda) + 2 \cdot 4 = 0$$

$$\Rightarrow \lambda^2 - 4\lambda + 11 = 0$$

$$e(A) = A^2 - 4A + 11 = 0$$

$$A^2 = A \cdot A = \begin{pmatrix} 1 & -2 \\ 4 & 3 \end{pmatrix}\begin{pmatrix} 1 & -2 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} -7 & -8 \\ 16 & 1 \end{pmatrix}$$

$$e(A) = C_0 I + C_1 A + C_2 A^2$$

$$= 11\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - 4\begin{pmatrix} 1 & -2 \\ 4 & 3 \end{pmatrix} + \begin{pmatrix} -7 & -8 \\ 16 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Ασκήσεις 8.3

(1) $2x^2 - xy + 5y^2$, $x^2 + y^2 = 1$

$A = \begin{pmatrix} 2 & -1/2 \\ 1/2 & 5 \end{pmatrix}$ $\det(A - \lambda I) = (2-\lambda)(5-\lambda) - \frac{1}{4} = 0 \Rightarrow 4\lambda^2 - 28\lambda + 39 = 0$

$\lambda_{1,2} = \frac{28 \pm \sqrt{28^2 - 4 \cdot 4 \cdot 39}}{2 \cdot 4}$ $\lambda^1 = \frac{7+\sqrt{10}}{2}$ (μέγιστο)

$\lambda^2 = \frac{7-\sqrt{10}}{2}$ (ελάχιστο)

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$$* \quad [A-\mu I \,|\, 0] = \left[\begin{array}{cc|c} \frac{-3-\sqrt{10}}{2} & -1/2 & 0 \\ -1/2 & \frac{3-\sqrt{10}}{2} & 0 \end{array}\right] \Rightarrow R_2 : (3+\sqrt{10})R_2 - R_1 \quad \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right]$$

$$R_1 : R_1 - R_2$$

$$R_1 = \frac{-2}{31\sqrt{10}} R_1$$

$$R_2 = -2R_2$$

Παράρτημα Α

Ασκήσεις Α.2

(1) e) $A = \begin{pmatrix} 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 4 \end{pmatrix}$, $A \in \mathbb{R}^{2\times 4} \Rightarrow A^T = A$

$$A \cdot A^T = \begin{pmatrix} 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 4 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 5 & 11 \\ 11 & 25 \end{pmatrix}$$

$$A^T \cdot A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 3 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 10 & 14 \\ 0 & 0 & 14 & 20 \end{pmatrix}$$

$e(\mu) = \det(AA^T - \mu I_2) = 0$

$$A \cdot A^T - \mu I_2 = \begin{pmatrix} 5 & 11 \\ 11 & 25 \end{pmatrix} - \mu \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 5-\mu & 11 \\ 11 & 25-\mu \end{pmatrix}$$

$e(\mu) = \det(A \cdot A^T - \mu I_2) = (5-\mu)(25-\mu) - 121 = 0 \quad \mu_{1,2} = 29,86 \;,\; 0,134$

$\Rightarrow \mu^2 - 30\mu + 4 = 0 \quad \mu_{1,2,3,4} = 29,86 \,|\, 0,134 \,|\, 0 \,|\, 0$

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$$U=\begin{pmatrix}0,4 & -0,94\\ 0,94 & 0,40\end{pmatrix} \qquad V=\begin{pmatrix}0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0,82 & 0,58\\ 0 & 0 & -0,58 & 0,82\end{pmatrix}$$

$$\Sigma=\begin{pmatrix}0 & 0\\ 6 & 0\\ 5,47 & 0\\ 0 & 0,37\end{pmatrix} \Rightarrow A=U\cdot\Sigma\cdot V$$

Κεφάλαιο 1

Ασκήσεις 1.4

(11) β) $=\frac{1}{32}(10-30\cos^2\theta+15+6\cos^4\theta-2\cos\theta\cos3\theta+\cos4\theta)$

$=\frac{1}{32}(25-30\cos^2\theta+(7-4\cos^2\theta)(-2\cos^2\theta+1)+2\cos\theta\cos3\theta(7-4\cos^2\theta+1))$

$=\frac{1}{32}(32-48\cos^2\theta+8\cos^4\theta+64\cos^4\theta-48\cos^2\theta+24\cos^4\theta-32\cos^6\theta)$

$=\frac{32}{32}(1-3\cos^2\theta+3\cos^4\theta-\cos^6\theta)$

$=(\sin^2\theta)^3 \Rightarrow$ Α' Μέλος

Ασκήσεις 2.3

(3) δ) $e^{2-\pi i}=1+i$ $\quad z=(x+yi)$

$\ln e^{2-\pi i}=\ln(1+i)$ $\quad x=0$, $\pi=i$

$2-\pi i=\ln(1+i)$

$2-\pi i=0$

ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ

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Ασκήσεις 2.4

(1) α) $\int_{\pi/4}^{\pi/2} te^{2it}dt = \int_{\pi/4}^{\pi/2} t(\cos 2t + i\sin 2t)dt = \int_{\pi/4}^{\pi/2} t\cos 2t\,dt + \int_{\pi/4}^{\pi/2} ti\sin 2t\,dt$

$u = t \Rightarrow du = dt$

$\int dv = \int \cos 2t\,dt$

$\Rightarrow v = \frac{\sin 2t}{2}$

$u = t \Rightarrow du = dt$

$\int dv = \int \sin 2t\,dt$

$\Rightarrow v = -\frac{\cos 2t}{2}$

$= \left[\frac{t}{2}\sin 2t\right]_{\pi/4}^{\pi/2} - \int_{\pi/4}^{\pi/2} \sin 2t\,dt - \left[\left[\frac{t}{2}\cos 2t\right]_{\pi/4}^{\pi/2} + \int_{\pi/4}^{\pi/2} \cos 2t\,dt\right] i$

$= \frac{\pi}{4}\sin\pi - \frac{\pi}{8}\sin\frac{\pi}{2} + \cos\frac{\pi}{4} - \frac{\cos\frac{\pi}{2}}{4} - \left[\frac{\pi}{4}\cos\pi + \frac{\pi}{8}\cos\frac{\pi}{2} + \sin\frac{\pi}{4} - \frac{\sin\frac{\pi}{2}}{4}\right] i$

$= -\frac{\pi - 2}{8} - i\left(\frac{-\pi - 4}{4}\right)$

Ασκήσεις 3.1

(1) α) $\int_{1+i}^{1+3i} z^2 dz = \left[\frac{z^3}{3}\right]_{1+i}^{1+3i} = \frac{(1+3i)^3}{3} - \frac{(1+i)^3}{3}$

$= \frac{(1+9i+27-27i)}{3} - \frac{(1+3i-3-i)}{3} = \frac{-24-20i}{3}$

Ασκήσεις 3.2

(2) α) $\oint \frac{1}{z^4+4} dz$, $C = C^+, (-1+i)$ $\quad Res[f, \sqrt{2}] = \lim (z+\sqrt{2}) f(z)$

$\oint_C \frac{1}{z^4+4} dz = 2\pi i \sum_{i=1}^{n} Res[C, z_i]$ $\quad \lim_{z\to\sqrt{2}} \frac{z+\sqrt{2}}{z^4+4} = \frac{2+\sqrt{2}}{-4+4} = \frac{0}{0}$

$f(z) = \frac{1}{z^4+4}$ $\quad z^4+4=0$ $\Rightarrow z^4 = -4$ $\quad z = -\sqrt{2}$

(16)

ΠΡΩΤΟΠΟΡΙΑ ΠΑΝΕΠΙΣΤΗΜΙΟ ΚΥΠΡΟΥ ΤΕΧΝΟΛΟΓΙΚΟΥ

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